

NUZHAT KHAN

SOFTWARE DEVELOPER | PYTHON • AI/GENAI • RAG • VECTOR SEARCH • LLM APPLICATIONS • FASTAPI

Mumbai, India | +91-9769149366 | knuzhat136@gmail.com | linkedin.com/in/nuzhat-khan-dev | github.com/Naazkn13 | portfolio

PROFESSIONAL SUMMARY

Software Developer specializing in Python, Generative AI applications, and RAG architectures. Experienced combining robust backend systems (FastAPI, PostgreSQL, Docker) with practical LLM workflows, vector databases (Qdrant, Pinecone), and multimodal embeddings (CLIP). Proven track record engineering complex document processing pipelines and rule-based enterprise software (Infomatics Services), bringing real-world data reliability into modern AI systems.

TECHNICAL SKILLS

AI & GenAI: RAG (Retrieval-Augmented Generation), Semantic Search, Embeddings, Vector Retrieval, LLM Integration, Prompt Engineering, LangChain

Vector Databases: Qdrant, Pinecone, FAISS, Similarity Search, Hybrid Search

Languages & Backend: Python, TypeScript, JavaScript, SQL, FastAPI, REST APIs, Pydantic, Microservices

Data Engineering: PyMuPDF, Tabula-py, Pandas, Unstructured Document Extraction, OCR & Parsing Pipelines

Databases & DevOps: PostgreSQL, Supabase, MongoDB, Docker, Git, GitHub Actions, Linux, CI/CD, Postman

Frontend & Deployment: React, Next.js, Tailwind CSS, Vercel, Railway, Azure DevOps

AI & ENGINEERING PROJECTS

FashionGallery — Multimodal Semantic Search & Discovery Engine — [GitHub](#)

Python • FastAPI • CLIP • Qdrant • RAG • Next.js • Docker

- **Multimodal Vector Retrieval:** Developed an AI-powered visual and textual discovery engine using OpenAI CLIP embeddings to compute joint image-text similarity vectors stored in Qdrant vector database.
- **Explainable RAG Layer:** Implemented RAG-style query expansion to retrieve relevant product metadata and explain why visual recommendations match user intent, overcoming traditional text-only search limits.
- **Production Architecture:** Designed asynchronous FastAPI endpoints serving similarity queries with Dockerized deployment for dev/prod parity.

CAS Parser — Unstructured Financial Document AI Extraction Pipeline — Client Deployed

Python • PyMuPDF • Tabula-py • Pandas • Tauri

- **Document Parsing at Scale:** Engineered an intelligent document pipeline converting semi-structured and unstructured CDSL statement PDFs into normalized tabular records with 98% extraction accuracy across 500+ statements/month.
- **High-ROI Automation:** Eliminated 10+ hours/week of manual document entry for financial operations teams by implementing deterministic layout extraction algorithms.

Biometric Attendance & Salary Engine — [attendance-sigma-one.vercel.app](#)

FastAPI • React • Supabase • Expo • ZKTeco SDK

- Deployed production full-stack platform managing real-time biometric event synchronization, configurable rule evaluation, and automatic payslip generation for healthcare staff, reducing payroll cycle duration by 75%.

PROFESSIONAL EXPERIENCE

Infomatics Services Pvt. Ltd. — Software Developer

Mumbai | June 2025 – 24 July 2026

- **Enterprise RegTech Platforms:** Shipped 4 production financial compliance platforms automating 67+ SEBI reporting templates and real-time trade pre-clearance across 4 financial data feeds (ECAS/BENPOS/KFin/CAMS).
- **Data & Workflow Governance:** Implemented 43 maker-checker stages, automated email event triggers, and tamper-resistant audit logs using FastAPI, PostgreSQL, and RBAC security models.
- **Engineering Standards:** Integrated Docker containerization and Azure DevOps CI/CD pipelines, ensuring production release reliability across institutional UAT deployments.

EDUCATION

BSc in Information Technology — Bhavan's College, Mumbai University

2021 – 2024

CERTIFICATIONS

Advanced Program in Java Full Stack Development — DVOC Institute (Skill India / NSDC)

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